RACHEL E

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EDUCATION

Bachelor of Technology in Computer Science Engineering with spec. in AI & ML
Periyar Maniammai Institute of Science and Technology, Thanjavur, Tamilnadu, India.
Current CGPA: 8.49/10 [until 5th semester]
Expected graduation: May, 2027.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, PHP, SQL, HTML/CSS
- **Machine Learning & Data Science:** Scikit-learn, Pandas, NumPy, Matplotlib, CatBoost
- **Developer Tools:** Git, Docker, Jupyter Notebook, Google Cloud Platform, VS Code
- **Core Coursework:** Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Organization & Architecture, Design and Analysis of Algorithms, Artificial Intelligence, Machine Learning, Probability & Statistics, Linear Algebra, Calculus, Numerical Computing, Algorithm Design, Data Analysis

EXPERIENCE

Research Intern **July 2025 - October 2025**
Council of Scientific and Industrial Research - 4th Paradigm Institute, Bengaluru, Karnataka, India.
Developed Job Recruitment Management System using PHP, HTML5, MySQL with Candidate Portal and custom built Job admin integrated into Joomla Content Management System for CSIR under the guidance of Dr.Anil Earnest, Principal Scientist at CSIR.

PROJECTS

CT Scan Reconstruction and Enhancement using Deep Learning

- Developed a CT scan enhancement and reconstruction system using PyTorch and OpenCV for improving low-quality medical scan images.
- Implemented denoising, CLAHE-based contrast enhancement, and Monte Carlo Dropout-based uncertainty estimation for confidence-aware reconstruction.
- Generated trust maps using edge detection and variance-based uncertainty analysis to visualize reconstruction reliability.
- Built an interactive Streamlit application for real-time CT image upload, reconstruction, and visualization.

Student exam score prediction using Machine Learning

- Developed and evaluated predictive machine learning models on a dataset of 20,000 student records to estimate academic performance.
- Conducted exploratory data analysis and statistical correlation analysis to identify feature interactions and nonlinear behavioural patterns.

- Implemented multiple regression-based and ensemble learning approaches including Random Forest, CatBoost, and ensemble models.
- Utilized Python libraries including Pandas, NumPy, Matplotlib, and Scikit-learn for data preprocessing, visualization, and model evaluation.

RESEARCH INTERESTS

Computational Engineering, Machine Learning, Computer Vision, Image Analysis, Algorithm Development, Intelligent Systems, Medical Image Processing, Deep Learning, Image Reconstruction, PyTorch.